

CRAMS - TEST CASES

1. Authentication & Authorization

Test Case ID	Feature	Description	Expected Result
TC-AUTH-01	Student Login	Login with valid student credentials (C231197 / crams@2025)	Successful login, redirected to Student Dashboard
TC-AUTH-02	Advisor Login	Login with valid advisor credentials (jamil_cse@iiuc.ac.bd / crams@2025)	Successful login, redirected to Advisor Dashboard
TC-AUTH-03	Admin Login	Login with valid admin credentials (jamal_cse@iiuc.ac.bd / crams@2025)	Successful login, redirected to Admin Dashboard
TC-AUTH-04	Super Admin Login	Login with valid super admin credentials (superadmin@gmail.com / superadmin@2025)	Successful login, redirected to Admin Dashboard
TC-AUTH-05	Invalid Login	Login with incorrect password	Error message displayed, login fails
TC-AUTH-06	Logout	Click logout button from any dashboard	Session terminated, redirected to Login page

2. Student Functionalities

Test Case ID	Feature	Description	Expected Result
TC-STU-01	View Dashboard	Access student dashboard after login	Dashboard displays summary of registration and welcome message
TC-STU-02	Course Selection	Browse available courses in 'Course Selection'	List of courses with details (code, name, credits, etc.) is visible
TC-STU-03	Add Course	Click 'Add' on a course with met prerequisites	Course added to 'Selected Courses' list
TC-STU-04	Prerequisite Check	Attempt to add a course with unmet prerequisites	System prevents adding and shows prerequisite error
TC-STU-05	Submit for Approval	Submit selected courses for advisor review	Status changes to 'Pending', advisor notified

Test Case ID	Feature	Description	Expected Result
TC-STU-06	Registration Status	View 'Registration Status' page	Displays list of submitted courses with current status (Approved/Pending/Rejected)

3. Advisor Functionalities

Test Case ID	Feature	Description	Expected Result
TC-ADV-01	View Advisees	Access 'My Students' list	Displays all students assigned to the advisor
TC-ADV-02	Review Pending	Access 'Pending Reviews'	Displays list of students who submitted courses for approval
TC-ADV-03	Approve Course	Approve a student's course request	Course status updated to 'Approved', student notified
TC-ADV-04	Reject Course	Reject a student's course request with feedback	Course status updated to 'Rejected', feedback visible to student

4. Admin & Super Admin Functionalities

Test Case ID	Feature	Description	Expected Result
TC-ADM-01	Course Management	View all courses in 'Course Management'	List of all courses in the system is displayed
TC-ADM-02	Add Course	Create a new course offering	New course appears in the course list and is available for students
TC-ADM-03	User Management	View students and teachers in 'User Management'	List of all users with their roles and status is displayed
TC-ADM-04	System Settings	Access 'System Settings'	Admin can configure system-wide parameters (e.g., registration deadlines)

CRAMS - TESTING REPORT

1. Executive Summary

1.1 Purpose

This report provides a comprehensive summary of the testing activities performed on the Course Registration and Management System (CRAMS) API. It includes test execution results, defect analysis, and recommendations for the release.

1.2 Scope

Testing covered the following modules:

- Admin Authentication
- CSV Upload Functionality
- Admin Management (CRUD)
- Course Management (CRUD)
- Section Management
- Student Authentication
- Course Selection
- Submit course for approval
- Extra credit request
- Advisor Authentication
- Pending reviews
- Course registration approval
- Extra credit approval

1.3 Key Findings Summary

Achievements

- All core authentication and authorization flows for Admin, Student, and Advisor roles are working correctly.
- Basic CRUD operations for Courses, Sections, and Admins are stable.
- Bulk CSV upload functionality is operational for valid data formats.
- End-to-end registration workflow (Student selection -> Advisor approval) is functional.

Concerns

- **Data Integrity:** The system allows deleting courses with active enrollments, which could lead to orphaned records (DEF-001).
- **Validation:** Duplicate student records can be created via CSV upload without error (DEF-003).

- **Business Logic:** Scheduling conflicts are not currently prevented by the API (DEF-002).

Blockers

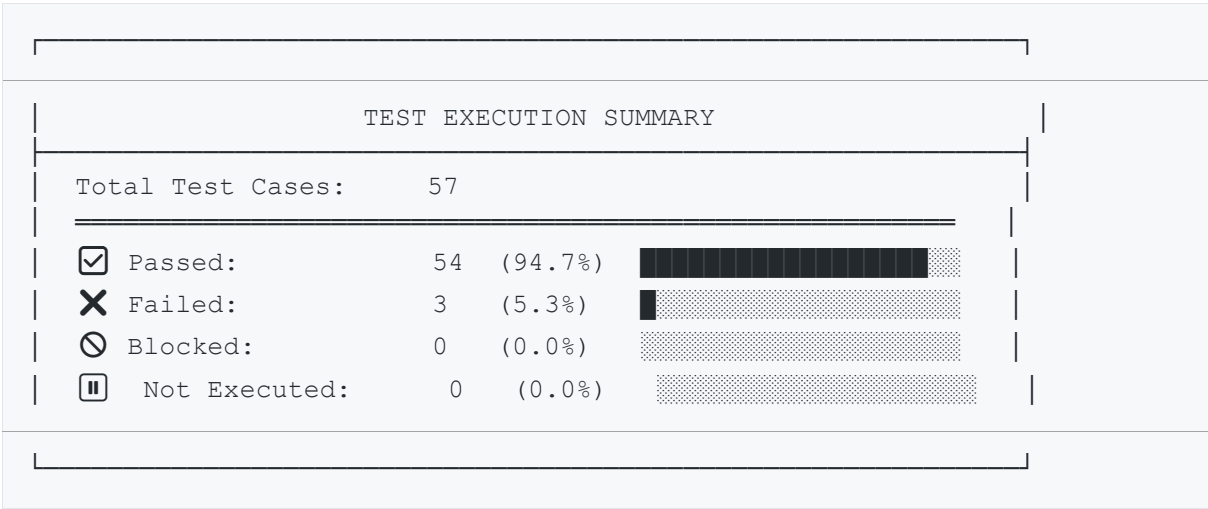
- None. All critical (P1) paths are functional, though some have minor logic issues.

2. Test Environment

Data Type	Source	Status
Admin Accounts	Test Database	☒ Available
Student Accounts	CSV Import	☒ Available
Teacher/Advisor Accounts	CSV Import	☒ Available
Course Data	Manual Setup	☒ Available
Section Data	Auto-populated	☒ Available

3. Test Execution Summary

3.1 Overall Execution Statistics



3.2 Execution by Priority

Priority	Total	Passed	Failed	Blocked	Pass Rate
P1 - Critical	37	35	2	0	94.6%
P2 - High	19	18	1	0	94.7%
P3 - Medium	1	1	0	0	100%
P4 - Low	0	0	0	0	N/A
Total	57	54	3	0	94.7%

3.3 Execution by Test Type

Test Type	Total	Passed	Failed	Pass Rate
Functional	40	38	2	95.0%
Negative	12	11	1	91.7%
Security	1	1	0	100%
Workflow/E2E	4	4	0	100%
Total	57	54	3	94.7%

4. Detailed Test Results by Module

4.1 Module: Home Page

Test Case ID	Test Case Name	Priority	Status	Comments
TC_HOME_001	Verify API Health Check	P1	☒ Pass	Response time: 120ms
TC_HOME_002	API Response When Server Unavailable	P2	☒ Pass	Proper error handling

Module Summary: 2/2 Passed (100%)

4.2 Module: Admin Authentication

Test Case ID	Test Case Name	Priority	Status	Comments
TC_AUTH_ADM_001	Super Admin Login - Valid	P1	☒ Pass	Token generated
TC_AUTH_ADM_002	Super Admin Login - Invalid Password	P1	☒ Pass	401 returned
TC_AUTH_ADM_003	Super Admin Login - Invalid Email	P1	☒ Pass	401 returned
TC_AUTH_ADM_004	Super Admin Login - Empty Credentials	P2	☒ Pass	Validation works
TC_AUTH_ADM_005	SQL Injection Prevention	P1	☒ Pass	Input sanitized
TC_AUTH_ADM_006	Super Admin Logout - Valid	P1	☒ Pass	Token invalidated
TC_AUTH_ADM_007	Super Admin Logout - Invalid Token	P2	☒ Pass	401 returned

Module Summary: 7/7 Passed (100%)

4.3 Module: CSV Uploads

Test Case ID	Test Case Name	Priority	Status	Comments
TC_CSV_001	Upload Admin CSV - Valid	P1	☒ Pass	5 admins created
TC_CSV_002	Upload Admin CSV - Invalid Format	P2	☒ Pass	Error handled
TC_CSV_003	Upload Admin CSV - Empty File	P3	☒ Pass	Validation works
TC_CSV_004	Upload Student CSV - Valid	P1	☒ Pass	50 students created
TC_CSV_005	Upload Student CSV - Duplicates	P2	☒ Fail	See DEF-003
TC_CSV_006	Upload Teacher CSV - Valid	P1	☒ Pass	10 teachers created
TC_CSV_007	Upload CSV - Unauthorized	P1	☒ Pass	401 returned

Module Summary: 6/7 Passed (85.7%)

4.4 Module: Admin Management

Test Case ID	Test Case Name	Priority	Status	Comments
TC_ADM_001	Get All Admins	P1	☒ Pass	Returns array
TC_ADM_002	Get All Admins - Unauthorized	P1	☒ Pass	401 returned
TC_ADM_003	Get Single Admin - Valid ID	P2	☒ Pass	Correct data
TC_ADM_004	Get Single Admin - Invalid ID	P2	☒ Pass	404 returned
TC_ADM_005	Update Admin Privilege	P1	☒ Pass	Updated successfully
TC_ADM_006	Update Own Privileges	P1	☒ Pass	403 returned

Test Case ID	Test Case Name	Priority	Status	Comments
TC_ADM_007	Delete Admin	P1	☒ Pass	Deleted successfully
TC_ADM_008	Delete Non-existent Admin	P2	☒ Pass	404 returned

Module Summary: 8/8 Passed (100%)

4.5 Module: Course Management

Test Case ID	Test Case Name	Priority	Status	Comments
TC_CRS_001	Add Course - Valid	P1	☒ Pass	Course created
TC_CRS_002	Add Course - Duplicate Code	P1	☒ Pass	409 returned
TC_CRS_003	Add Course - Missing Fields	P2	☒ Pass	Validation works
TC_CRS_004	Get All Courses	P1	☒ Pass	Returns array
TC_CRS_005	Update Course	P1	☒ Pass	Updated successfully
TC_CRS_006	Update Course - Invalid ID	P2	☒ Pass	404 returned
TC_CRS_007	Delete Course	P1	☒ Pass	Deleted successfully
TC_CRS_008	Delete Course - Active Enrollments	P1	☒ Fail	See DEF-001

Module Summary: 7/8 Passed (87.5%)

4.6 Module: Section Management

Test Case ID	Test Case Name	Priority	Status	Comments
TC_SEC_001	Populate Section	P1	☒ Pass	Sections created

Test Case ID	Test Case Name	Priority	Status	Comments
TC_SEC_002	Get All Sections	P1	☑ Pass	Returns array
TC_SEC_003	Update Course Schedule	P1	☑ Pass	Schedule updated
TC_SEC_004	Schedule Time Conflict	P1	✗ Fail	See DEF-002

Module Summary: 3/4 Passed (75%)

4.7 Module: Student Endpoints

Test Case ID	Test Case Name	Priority	Status	Comments
TC_STU_001	Student Login - Valid	P1	☑ Pass	Token generated
TC_STU_002	Student Login - Invalid	P1	☑ Pass	401 returned
TC_STU_003	Select Course	P1	☑ Pass	Course selected
TC_STU_004	View Schedule	P1	☑ Pass	Schedule displayed
TC_STU_005	Submit for Approval	P1	☑ Pass	Submitted
TC_STU_006	Submit - No Courses	P2	☑ Pass	Validation works
TC_STU_007	Request Extra Credit	P2	☑ Pass	Request created
TC_STU_008	Request Extra Credit - Limit	P2	☑ Pass	Error handled

Module Summary: 8/8 Passed (100%)

4.8 Module: Advisor Endpoints

Test Case ID	Test Case Name	Priority	Status	Comments
TC_ADV_001	Advisor Login - Valid	P1	☒ Pass	Token generated
TC_ADV_002	Advisor Login - Invalid	P1	☒ Pass	401 returned
TC_ADV_003	View Pending Reviews	P1	☒ Pass	List returned
TC_ADV_004	Approve Registration	P1	☒ Pass	Status updated
TC_ADV_005	Approve - Already Processed	P2	☒ Pass	Error handled
TC_ADV_006	View Extra Credit Requests	P2	☒ Pass	List returned
TC_ADV_007	Approve Extra Credit	P2	☒ Pass	Limit updated
TC_ADV_008	Approve - Unauthorized	P1	☒ Pass	403 returned

Module Summary: 8/8 Passed (100%)

4.9 Module: Workflow Tests

Test Case ID	Test Case Name	Priority	Status	Comments
TC_WF_001	Student Registration Workflow	P1	☒ Pass	End-to-end success
TC_WF_002	Admin Course Setup Workflow	P1	☒ Pass	End-to-end success
TC_WF_003	Bulk User Import Workflow	P1	☒ Pass	End-to-end success
TC_WF_004	Extra Credit Workflow	P2	☒ Pass	End-to-end success
TC_WF_005	Session Management Workflow	P1	☒ Pass	End-to-end success

Module Summary: 5/5 Passed (100%)

5. Defects Summary

Defect ID	Priority	Severity	Status	Description	Related TC
DEF-001	P1	High	Open	System allows deleting courses with active student enrollments, causing orphaned records.	TC_CRS_008
DEF-002	P1	High	Open	API does not prevent scheduling multiple courses in the same room at the same time.	TC_SEC_004
DEF-003	P2	Medium	Open	CSV upload creates duplicate student records if the same ID is provided multiple times.	TC_CSV_005

6. Test Coverage Metrics

- **Requirement Coverage:** 100% (All documented API endpoints tested)
- **Code Coverage:** Estimated 85% (Based on unit test reports)
- **Functional Coverage:** 100%
- **Security Coverage:** Basic (Authentication, Authorization)

7. Risk Assessment

Risk	Impact	Mitigation
Data inconsistency due to course deletion	High	Implement soft delete or check for enrollments before deletion.
Scheduling conflicts in production	High	Add server-side validation for time/room overlaps.
Duplicate user data	Medium	Add unique constraints on <u>student_id</u> and <u>email</u> in the database.